
Project: Modelling Human Travel Behavior with GPS tracking data

Background

GPS tracking has been conducted for the purpose of improving the quality and the quantity of travel data in many scenarios. It could generate detailed and accurate documentation of travel movements.

A data collection has recruited 21 volunteers with their proper home address to track their movements. The whole data collection lasted from March to September in 2017.

The phone application activated tracking every 5 or 30 seconds. The recorded information included Date, time, longitude, latitude, and Speed. There were 369, 940 valid positional recordings.

The purpose of this project is to conduct spatial data analysis from the basic part: data processing, create features from the data, project the coordinate systems, selection, attribute table, etc. to the advanced part, redefine data, data accuracy, pattern analysis and so on.

You need to finish your assignment by going through the following tasks. **The mobility data is in a folder named by Mobility data set.** Find the data under the folder in Learn. In addition, you have another data set “**Data for all**” that is road network data and Sweden municipality. If you have any questions regarding the dataset, please contact me psd@du.se

Specific Objectives and Aim

The aim of this project is to investigate the human mobility pattern analyzing how the mobility patterns are, and how much people use different transport modes during daily life. The objectives of the project are to:

- Learn how to work with the large dataset;
- Preparing the Raw GPS data;
- Heat map and pattern analysis;
- Network analysis
- Working with the dataset and visualization of the data at the same time;
- Find a pattern for the mobility of humans and predict their future location.

Task 1: Process the data

Prepare the dataset for QGIS by processing GPS data according to specific criteria and calculations. The final data should be saved in either CSV or XLSX format.

- Generate Identifiers:
 1. Trip ID: Create a new ID for each trip. If the time difference between consecutive data points exceeds 180 seconds, start a new trip ID.
 2. Point ID: Assign a unique ID to each point within its respective trip.
- Filter Data:
 1. Speed Limit: Exclude any data points where the speed exceeds 200 km/h.
- Distance Calculation:
 1. Calculate the geographical distance between each consecutive pair of points.
- Time Difference Calculation:
 1. Determine the time difference between consecutive points.
- Speed Calculations:
 1. Average Speed per Trip: Calculate the average speed for each trip by averaging the speeds of all points within the trip.
- Acceleration Calculation:
 1. Calculate the acceleration at each point using the change in speed over time between consecutive points.
 2. Average Acceleration per Trip: Calculate the average acceleration for each trip.
- Trip Summary Statistics:
 1. Duration: Calculate the total duration for each trip.
 2. Distance: Calculate the total distance traveled in each trip.
- Ensure all calculations and formatting comply with the specified requirements.
- Save the processed dataset in CSV or XLSX format, ready for upload to QGIS.

Task 2: Create the layers

In this task, you will create new layers using the pre-processed GPS measurements, as well as existing layers.

1. After you pre-processed data originating from GPS measurement of volunteers, add these data into QGIS. Given that your data are saved as XLSX file, use function **Layer > Add spreadsheet layer**. Keep in mind that this time you need to create a layer with geometry; thus, fill in the option **Geometry** and select in which fields in the Excel sheet there are recorded X and Y coordinates, respectively. **The reference system** is WGS84.
2. Remove all positional recording outside of Sweden boundaries.
3. In order to make the *spreadsheet* layer permanent, right-click the layer name in the panel Layers and select **Export > Save features as**. Label the layer as “Positional recordings” At this point, and you have created a point layer showing GPS measurements.
4. As the *background* layer for the analysis, you will use a layer of Sweden road network. Add vector layer the Sweden_Roads_Networks to your layers.
5. As the *background* layer for the analysis, you will use a layer of **Falun municipality**. This layer you need to select from the layer **Kommuner_Sverige** and export into a new layer labeled **Falun municipality**.

At the end of this task, you should be able to set a source coordinate system for the layer, and the selection.

Output:

The output for this task should be a map with all layers with title, legend, scale bar, and north arrow for the map.

Task 3: Network Analysis

Creating, visualizing, and managing networks is an important part of GIS. Many types of physical infrastructure such as roads, railways, utilities can be modeled as networks with lines and nodes - with properties attached to them. In this task we work with how to make path from positional recording data and how to analysis them.

1. Using the point layer “Positional recording”, create a line layer showing volunteers’ tracks. Generate tracks using tool **Point to path**. To launch this tool, select from the main toolbar **Processing > Toolbox**. A new panel with **Toolbox** should appear in the right part of the QGIS. Search for the tool by typing **Point to path** to the upper window. Label the layer “Tracks”.

2. Try to categorize the tracks by Driver_ID and show each Driver_ID by specific color, based on your output can you see any pattern?
3. The shortest path suggested by Google Maps is based on shortest path algorithm, by doing this subtask you can understand how google maps works to give the costumer suggestion. Separate and create another excel file for one track from your dataset. In toolbox, search for path to path (point to point) dialog. select Sweden_Road_Networks as the Vector layer representing network. Keep the Path type to calculate as Shortest. Next, we need to pick a start and end point. You can click the button and click on the first and end point  on your sample track the network.
 - a. Try to calculate the shortest path for one track and compare with real mobility behavior. Also, compare the shortest path result with google map suggestions.

Output:

At the end of this task, you should be able to create path from point dataset and analysis the path.

Task 4: Pattern analysis

A heat map is a graphical representation of point data that is used to visualize the diversity of data. A heat map is the best tool to understand where are a high concentration of point data and the cluster. In addition, this tool use for Hotspot analysis; it has a wide variety of applications, for example, the crime hotspot, traffic clustering, or any data concentration. Now you have all your layers ready. You conduct a point pattern analysis for your Positional recordings. [Note: It might take very long to run the pattern analysis; try not to use all your data at once, select a small example to test if it works. If the whole data set does not work, you can show your results with a subset of your data.] You can find a tutorial of analysis in this hyperlinks:

https://www.qgistutorials.com/en/docs/3/creating_heatmaps.html

1. Heatmaps are one of the best visualization tools for dense point data. heatmap is an interpolation technique useful in determining the density of input features. Heatmaps are most commonly used to visualize crime data, traffic incidents, housing density, etc. QGIS has a heatmap renderer that can be used to style a point layer and a Processing algorithm Heatmap (Kernel Density Estimation) that can be used to create a raster from a point layer. Generate your heatmap based on your positional recording data (point data). In order to make a heatmap, you need right click on your positional recording data and click on properties, on Symbology tap change the simple symbol to the heatmap and define color ramp for your heatmap.
 - a. Explain your results and show which area based on your dataset in Borlänge has more density of positional recording
2. Point cluster is another type of pattern analysis, and Point cluster renderers are used frequently in web mapping applications to condense multiple overlapping or nearby points into a single rendered marker for clarity. you can find a tutorial of this analysis

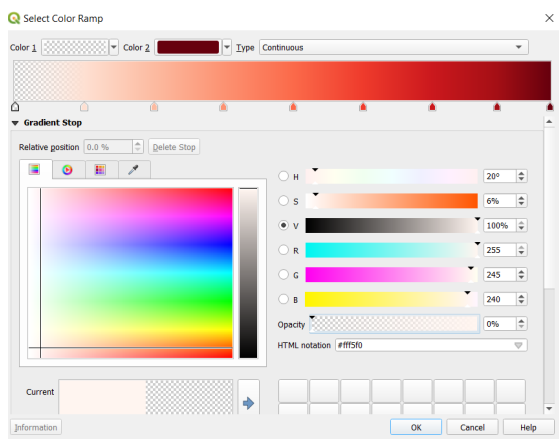
in this hyperlinks:

<https://north-road.com/qgis-cluster-renderer-crowdfunding/>

- a. Try to set a point cluster analysis, and based on your result, explain which part of the city has a high dense of positional recording data.
3. Point displacement also is one of the cluster analyses; in order to use this method, you need to define the placement method; you can make a choice between ring, grid, and concentric rings. Also, you can find relate tutorial in this hyperlink:
https://docs.qgis.org/2.14/en/docs/user_manual/working_with_vector/vector_properties.html#point-displacement
 - a. Define a point displacement analysis on your dataset (positional recording) and try to explain what your map is showing.
4. Put all map in one figure, can you see any difference between these three maps?

Hint:

- 1) When you set a color for your heatmap, define the first color as no color, for doing this you need to change the opacity to 0%. After doing this, you can have a better view of your heatmap.



- 2) After this task is finished, you need to change it again to a single symbol and works on other tasks.

Output:

The output for this task should be proper maps for each step (four maps in the end), and your explanation about your result. The objective of this task is how you can do the statistical analysis.

Task 5: Speed

Objective: Use recorded GPS data to generate a thematic map illustrating speed variations. Classify each positional recording into one of seven speed categories to better understand traffic flow patterns.

Speed Categories:

- Below 10 km/h
- 10 km/h to 20 km/h
- 20 km/h to 30 km/h
- 30 km/h to 50 km/h
- 50 km/h to 70 km/h
- 70 km/h to 90 km/h
- More than 90 km/h

Based on this task you can have a better view of traffic flow in your dataset.

Create Speed Variation Map:

Data Representation: Each point on the map should be colored according to its speed category, with distinct colors representing each range.

Coverage: The map should display the entire dataset, showing how speed distributions vary across different areas.

Zoom-In: Select a specific road or area for a detailed view. This zoomed-in map should highlight particular traffic flow characteristics or notable speed patterns.

Task 5: Gender-Based Heatmap Creation

Objective: Develop thematic maps to visualize differences in speed behavior between genders using recorded GPS data. The dataset should be categorized into two groups, Female and Male, for comparative analysis. **Identify Gender:** Use 'F' to represent Female and 'M' to represent Male in your dataset.

Group Data: Separate the data into two distinct groups based on the gender field.

Create Heatmaps:

Male Heatmap: Develop a heatmap using the data classified under 'M'. This map should visually represent the speed and frequency of travel for male participants across the entire dataset.

Female Heatmap: Similarly, create a heatmap for data classified under 'F', representing female participants.

Output

Save and submit the two heatmaps as separate files.

Provide a written report or analysis section detailing the observations and interpretations derived from the heatmaps.

Task 6: Create your own

Based on your understanding of the data, generate at least one map (or more than one if it is needed) to illustrate the part that you think is representative and informative to show the movements of the volunteers. Basically, it could be anything, but first, it should be an interesting question that is related to the spatial analysis, and second, it depends on to whom and to where your analysis result(s) will be used, which means, the main purpose/usage of your map and analysis. For example, it could be a visualization of mobility patterns, some problematic parts that you uncovered during your procedure, so on.

- Clustering analysis
- Location problem
- Origin and Destination analysis
- Shortest path
- Machine learning algorithms

This final task is a reflection on how you have understood the spatial data analysis and GIS; you should think carefully and show in detail in your report.

Seminar Submission and Presentation Instructions

To ensure that your seminar submission and presentation are clear and effectively communicate your findings, here is a refined set of instructions:

Report Structure:

Introduction: Outline the objectives and scope of your analysis.

Methodology: Describe the procedures used in data analysis and map creation.

Results: Present the findings in a structured format. Include maps and other relevant figures.

Analysis: Discuss the implications and insights derived from the results.

Figure Requirements:

All maps must include a north arrow, scale bar, and a clear legend.

Ensure that all figures are created directly from your analysis software; screenshots are not acceptable.

Explanation and Clarity:

Provide thorough and clear explanations of all procedures, results, and analyses.

Make sure the report is well-written and free of grammatical errors.

Report Formatting:

Include your name in the filename (e.g., pariasadeghian.pdf) to personalize your submission.

Submission:

Submit your report in PDF format to ensure consistency and accessibility.

Presentation Preparation:

Presentation Content:

Create a presentation that summarizes your report, focusing on key results and insights.
Prepare to highlight major findings and explain the significance of your maps and data analysis.

Presentation Duration:

You have 5-10 minutes to present your results.

An additional 5 minutes will be allocated for answering questions from the audience.

Visual Aids:

Use clear and informative slides to aid your presentation.

Include relevant figures and excerpts from your report to illustrate your points effectively.

Additional Information:

Data Usage: The data provided can only be used for this specific course project.

Contact Information: For any queries related to your project or submission, you can contact psd@du.se.

Material:

All material for this tasks can be found in Assignment 2.zip in Canvas.

Deadline:

Assignment 2 needs to be submitted by 26/5 by 23:59 in Canvas.

Grade:

This assignment is graded as U/G/VG.

Note: In this task you can work individually or in groups of 3.